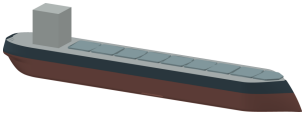
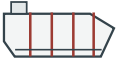
Design space



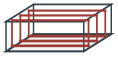
Early-stage bulk carrier



Hull dimensions



Compartments



Bottom structure



Side shell

189 parameters · 133 controls

20 subsystem blocks

Dataset size

100,000 random designs

+ 117,999 design repairs

System Evaluation:
187 Constraints
5 Objectives

170,439

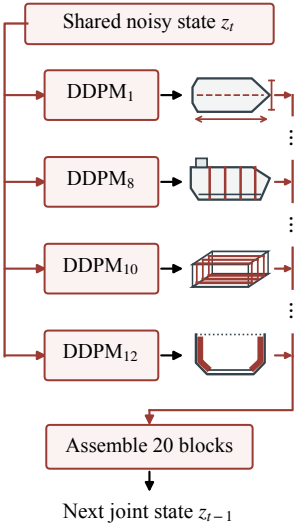
completed evaluations

☑ **16,247 fully feasible**

Parent-grouped data splits

Model architecture

SCDDPM · one reverse step



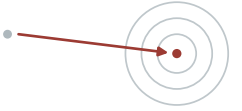
Sampling features

Feasibility guidance



feasible region

Performance guidance



lower RFR

Subsystem redesign



Hold selected blocks fixed